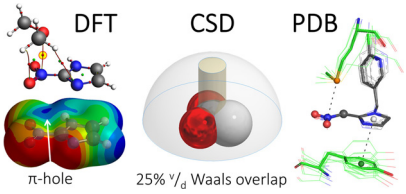
Synthesis J. M. Hoffmann et al. Paper

π-Hole Interactions with Various Nitro Compounds Relevant for Medicine: DFT Calculations and Surveys of the Cambridge Structural Database (CSD) and the Protein Data Bank (PDB)

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Abstract Model DFT computations and a thorough evaluation of the Cambridge structural database (CSD) and the protein data bank (PDB) were conducted to assess the occurrence and significance of intermolecular π-hole interactions with various nitro compounds relevant to medicine. DFT calculations indicate interaction energies between –3.9 to –6.5 kcal·mol^{-1}, which is in the order of typical hydrogen- and halogen-bonding interactions. Ample structural evidence for the occurrence of nitro π-hole interactions was found within the CSD and the PDB.

Key words hydrogen bonding, halogen bonding, interaction energy, van der Waals overlap, crystal sructure, protein structure

1 Introduction

Phenomena of molecular binding, such as ligand–protein interactions, are largely programmed by non-covalent forces.[1] Hydrogen-bonding[2] and more recently also halogen-bonding[3] interactions are well-known biologically relevant intermolecular interactions.[4] Both have been classified as so-called σ-hole interactions. A σ-hole can be seen as an electropositive potential located on a non-metal atom[5] along the vector of a covalent bond.[6] Similarly, electron-deficient π-systems such as carbonyls[7] and π-acidic aromatics[8] have been classified as so-called π-holes.[6c,9] These π-holes can interact favorably with electron-rich entities like lone-pair or π-electrons.[10]

A π-hole can also be found on nitro derivatives (R–NO_2) and is typically located above/below the N-atom or somewhere in between the R–N bond.[11] The potential of this π-hole depends largely on the nature of R, but can always be amplified when electron density is withdrawn from the O-atoms.[6c,11a,b,12] Indeed, even nitrate anions can bear a π-hole when the negative charge is sufficiently spread out over a larger area through hydrogen bonding[13] or coordination to a metal.[14]

Nitrate esters[11b] and nitro aromatics[12] have been highlighted as particularly interesting nitro compounds as they are found as ligands in protein structures and are used in important medicine. For example, isosorbide dinitrate is commonly prescribed to treat chronic congestive heart failure[15] and chloramphenicol is an inexpensive and abundantly used antibiotic;[16] both are listed as an essential medicine by the world health organization (WHO).[17]

It was thus wondered if there are other small nitro-containing molecules that engage in π-hole like interactions and whether such interactions could be relevant in medicine. This indeed seems to be the case, as detailed in the combined literature, density functional theory (DFT) and crystal structure database study reported herein.

2 Materials and Methods

2.1 Density Functional Theory (DFT) Computations

DFT geometry optimization calculations were performed with Spartan 2016 at the B3LYP[18]-D3[19]/def2-TZVP[20] level of theory, which is known to give accurate results at reasonable computational cost and a very low basis set superposition error (BSSE).[20] The geometry optimizations were performed without any constraints. The Molecular Electrostatic Potential Surfaces (MEPs) calculated are an aid to predict which regions of a molecule may act as electron-donor (red, negative) or -acceptor site (blue, positive). The Amsterdam Density Functional (ADF)[21] modeling suite at the B3LYP[18]-D3[19]/TZ2P[20] level of theory (no frozen cores) was used for energy decomposition and 'atoms in molecules' (AIM)[22] analyses.

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2.2 Analysis of the Cambridge Structure Database (CSD)

The CSD[23] version 5.40 including two updates (until May 2019) was inspected using ConQuest[24] version 1.23 (build 205818) and all searches were limited to single crystal X-ray structures that contained 3D coordinates and had an R-factor ≤0.1. As is depicted in the left hand side of Figure 1, data from the CSD were retrieved containing a nitro compound, XNO_2, where X can be any atom but not a methyl, aryl, OR, or O^- (i.e., nitrate ester or anion).

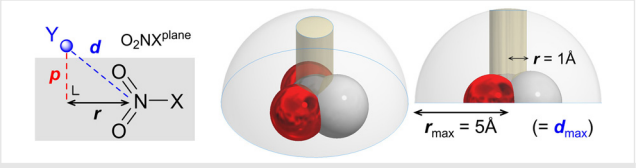


Figure 1 Illustration of the method used to assess the data in the CSD. Left: query used to retrieve data form the CSD. Y = an electron rich atom (N, P, As, O, S, Se, Te, F, Cl, Br, I, or At) or the H-atom of any methyl group. The intermolecular Y···N distance ***d*** ≤5Å, ***p*** is the distance between Y and the O_2NX plane, and ***r*** was calculated from ***d*** and ***p*** using Pythagoras' theorem. X can be any atom, but X = aryl, methyl, OR, or O^- (i.e., a nitrate ester or anion) was specifically excluded from this search. Middle and right: illustration of the 5Å high and wide hemisphere centered on an O_2NX model shown in space filling mode. The transparent golden body contains the data with ***r*** ≤1Å that was used to generated the plots in Figure 4 (vide infra) and Figure S4 [See Supporting Information (SI)].

Y was specified as either any electron rich atom (ElR: N, P, As, O, S, Se, Te, F, Cl, Br, I, or At) or the H-atom of any methyl group and in both cases the intermolecular Y···N distance ***d*** was restricted to ≤5Å. These data are thus confined within a hemisphere of 5Å height, centered on the nitro N-atom, as is illustrated in the middle and right of Figure 1 [for Y = ElR, *N* = 132.471 in 7.142 CIFs and for Y = H (methyl), *N* = 19.953 in 3.771 CIFs]. The absolute Y···O_2NX plane distance ***p*** was also measured, which allowed computation of the parallel displacement parameter ***r*** (Pythagoras' theorem). Data confined by ***d*** ≤5Å and ***r*** ≤1Å are thus confined within the transparent gold bodies shown in Figure 1. The relative amount of data (*N*) within these bodies was plotted as a function of the van der Waals corrected Y···N distance ***d'*** (i.e.: ***d***–vdW N–vdW Y).[25] The resulting *N*(***d'***) plots are shown in Figure 4 and Figure S4, respectively, for the data with Y = electron-rich atom and Y = H of any methyl.

2.3 Analysis of the Protein Data Bank (PDB)

The protein data bank (PDB)[26] was inspected manually using the ligand search option of the RSCB webpage (http://www.rcsb.org/) and searching for all general structures listed in Table S1 (see SI).

3 Results and Discussion

3.1 Medicinal Nitro-Containing Compounds and Model DFT Calculation

Besides nitrate esters and nitro aromatics[27] there are close to 60,000 small nitro-containing molecules that have been studied for their pharmacological properties.[28] As is detailed in entries **1**–**7** of Table S1, 48 such molecules were identified as approved by the U.S. Federal Drug Administration.[29] Eight of these are listed by the WHO as essential medicine[17] and only one involved an aliphatic nitro compound.[30,31] As is shown in the top of Figure 2 (and also in Table S1), the other approved drugs are basically nitroethene derivatives that can be divided into furans **1**, imidazoles **2**–**4**, thiazoles **5**, and ethene-1,1-diamines **6**. Examples of these compound classes (and their entry number in Table S1) include: Nitrofurantoin (entry **1**.5) used to lower urinary tract infections;[17,32] Nifurtimox (entry **1**.11) used to treat 2nd stage African trypanosomiasis (Chagan Disease) and *Trypanosoma brucei gambiense* infection;[17,33] Fexinidazole (entry **2**.5) for the treatment of 1st and 2nd stage of human African trypanosomiasis;[17,34] the widely used antibiotic Metronidazole (or Flagyl, entry **2**.7);[17,35] Delamanid (entry **3**.3) used as medicine for the treatment of multidrug-resistant tuberculosis;[17,36] Azathioprine (entry **3**.4) used as an immunomodulator for non-malignant diseases, for example as an anti-infectant during bowel surgery;[17,37] Benznidazole (entry **4**) is used to treat American trypanosomiasis (Chagan Disease);[17,33] Nitazoxanide (or Alinia, entry **5**.6) is an antiparacitic used to treat diarrhea;[38] and Ranitidine (or Zantac, entry **6**.1), which is used to treat *Helicobacter pylori* infection, a common cause of peptic ulcers.[17,39]

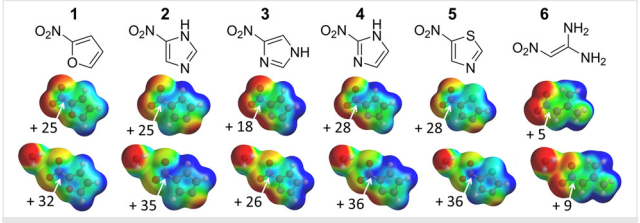


Figure 2 Overview of nitro-containing molecules **1**–**6** (top) and their electrostatic potential map (MEP, middle). The bottom figures are the MEPs of geometry optimized water adducts of **1**–**6**. Color codes from –27 kcal·mol^{-1} (red) to +36 kcal·mol^{-1} (blue) were computed at the DFT/B3LYP-D3/def2-TZVP level of theory. The numbers accompanying the MEPs are the maximum potentials located on the π-hole region and are usually predictive of the electrostatic component of an interaction.

The molecular electrostatic potential maps (MEPs) of model compounds **1**–**6** and their water adducts are shown in the middle and bottom of Figure 2, respectively. MEPs can be used to predict the location of a π-hole, as visualized by a more positive value (blue). The potentials of the π-holes range from +5 kcal·mol^{-1} in **6** to +28 kcal·mol^{-1} in **4**

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and **5**. The magnitude of these potentials is predictive of the strength of the electrostatic attraction to electron-rich entities (e.g., lone-pair or π electrons) that contribute to the total binding energy. Moreover, these potentials are in the order of the maximum electrostatic potentials on methylamine (+25 kcal·mol^{-1}) and molecular bromine (+31 kcal·mol^{-1}), which are prototypical hydrogen- and halogen-bond donors respectively.[11e] The potentials of the π-holes in **1–6** increase significantly when the nitro moiety is H-bonded to a water molecule with a maximum increase of 10 kcal·mol^{-1} in **2** (from +25 to +35 kcal·mol^{-1}). This is important because the polar groups of ligands are often hydrogen-bonded to water molecules or protein amides. For example, the potential of the π-hole on nitrobenzene itself is merely +13 kcal·mol^{-1},[11a,12] but this is amplified to +19 kcal·mol^{-1} in its water adduct and to +30 kcal·mol^{-1} when nitrobenzene is H-bonded to two amides.[12] The π-hole interactions of (hydrogen-bonded) nitroarene ligands have recently been reported to bear structural and functional significance in protein structures.[12]

The MEPs shown in Figure 2 thus predict an interaction with electron-rich entities such as lone-pair electrons when they are located above/below the nitro N-atom or the C–N bond. Shown in Table 1 (entries 1–6) is a numerical overview of computational data obtained from geometry optimized adducts of **1–6** with dimethyl ether (dme, entry a) and dimethyl thioether (dmt, entry b). Table 1 lists the intermolecular interaction energy (ΔE in kcal·mol^{-1}) and a decomposition analysis (in %) of the attractive electrostatic (E), orbital (O), and dispersive (D) interactions. An 'atoms-in-molecules' (AIM) analysis was also performed and Table 1 lists the density (ρ in 10^2 a.u.) of a bond critical point (bcp) between the O/S-atom in dme/dmt and a specified atom (X) of **1–6**.

The interaction energies of adducts with **1–6** are all favorable and vary between –6.5 kcal·mol^{-1} in [**1**···SMe_2] (Table 1, entry 1b) and –3.9 kcal·mol^{-1} in [**2**···OMe_2] (entry 2a). Except with **4**, the adducts with dmt (entry b) are slightly more stable than those computed with dme (entry a). This might be due to differences in polarizability (4.9 vs 19.3 a.u.), electronegativity (3.44 vs 2.58 on the Pauling scale), and/or the size (1.52 vs 1.80Å) of O versus S, or have to do with the different molecular volumes of dme (61 Å^3) and dmt (71 Å^3) resulting from the van der Waals radii of O/S and the C–O/S bond distances (1.41 vs 1.82 Å). The orbital contribution is rather constant at about 15–20% while the electrostatic and dispersive components each range between about 30–50%. As is also illustrated in Figure 3, The AIM analyses of the [**1–6**···OMe_2] adducts revealed a clear bcp between the O-atom of dme and the N-atom of the nitro moiety (with **1**, **2**, **4–6**) or the C-atom direct attached to NO_2 (with **3**) with a bond density between 0.007 and 0.011 a.u. This is similar for the dmt adducts (see Figure S3).

Table 1 Numerical Overview of Adducts Computed with DFT between Dimethyl Ether (Entry a) or Dimethyl Thioether (Entry b) as Lewis Base and Models **1–6** (Entries 1–6) or Reference Hydrogen/Halogen Bond Donors (Entries 9–12) as Lewis Acids[a]

Entry	Adduct	ΔE[b]	E/O/D[c]	ρ (X)[d]
1a	[**1**···OMe_2]	–5.3	44/16/40[e]	0.87 (**N**)
1b	[**1**···SMe_2]	–6.5	44/17/39	0.83 (**N**)
2a	[**2**···OMe_2]	–3.9	35/16/49	0.89 (**N**)
2b	[**2**···SMe_2]	–4.4	37/18/44	0.50 (**C**)
3a	[**3**···OMe_2]	–4.1	41/17/42	1.07 (**C**)
3b	[**3**···SMe_2]	–5.9	46/16/39	0.82 (**C**)
4a	[**4**···OMe_2]	–6.3	50/15/34	1.07 (**N**)
4b	[**4**···SMe_2]	–5.8	48/17/39	0.64 (**N**)
5a	[**5**···OMe_2]	–4.7	37/17/46	0.88 (**N**)
5b	[**5**···SMe_2]	–4.9	32/17/51	0.63 (**C**)
6a[f]	[**6**···OMe_2]	–4.5	36/17/47	0.65 (**N**)
6b[f]	[**6**···SMe_2]	–6.3	36/15/49	0.58 (**C**)
9a[f]	[OH_2···OMe_2]	–6.7	60/30/11	3.02 (H)
9b[f]	[OH_2···SMe_2]	–5.9	53/32/15	1.98 (H)
10a	[nma[g]-NH···OMe_2]	–7.1	56/24/20	2.39 (H)
10b	[nma[g]-NH···SMe_2]	–5.9	47/28/25	1.64 (H)
11a	[C_6H_5I···OMe_2]	–3.5	54/23/24	1.36 (I)
11b	[C_6H_5I···SMe_2]	–4.1	47/27/26	1.09 (I)
12a	[C_6F_5I···OMe_2]	–5.6	59/26/15	1.86 (I)
12b	[C_6F_5I···SMe_2]	–6.3	53/34/13	1.94 (I)

[a] See Figure 3 (vide infra) (and Figure S1, see SI) for perspective view of adducts with dme, Figure S2 (see SI) for adducts with dmt, and Figure S3 (see SI) for perspective views of the other adducts.
[b] Interaction energy in kcal·mol^{-1} of optimized adduct minus the energies of optimized molecules.
[c] Energy decomposition analysis of the total stabilization energy split up as percentages into electrostatics (E), orbital interactions (O), and dispersion (D).
[d] Density ρ of the bond critical point (in 10^2 arbitrary units) between O/S of the ether and an atom X in **1–6** (i.e., N of the nitro group or another atom).
[e] Recalculating the energies with different dispersion corrections led to nearly identical energies and energy contributions. That is, ΔE (% E/O/D) = –5.3 (44/16/40) @ D3; –5.5 (43/16/41) @ D; –4.7 (46/17/37) @ dUFF; and –4.8 (45/17/37) @ D3-BJ.
[f] Note that in this table entries 7 and 8 are not used to avoid confusion with entries 7 and 8 in Table S1.
[g] *N*-Methylacetamide.

For comparison purposes, dme/dmt adducts were also computed with the prototypical H-bond donors water (Table 1, entry 9) and *N*-methylacetamide (nma, entry 10), and with the typical halogen-bond donors iodobenzene (entry 11) and perfluoroiodobenzene (entries 12). The interaction energies vary between –7.1 kcal·mol^{-1} in [nma···OMe_2] (entry 10a, the strongest hydrogen bond calculated) and –3.5 kcal·mol^{-1} in [iodobenzene···OMe_2] (entry 11a, the weakest halogen bond evaluated). The π-hole complexes (entries 1–6) fall within this range. The electrostatic contribution in hydrogen- and halogen-bonded adducts is always largest

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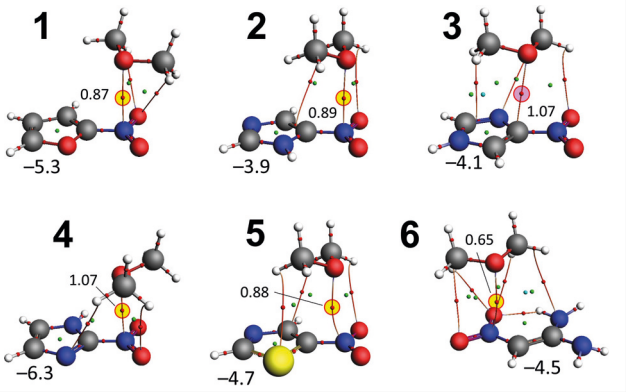


Figure 3 Perspective views of DFT optimized dimethyl ether (dme) adducts with **1–6** in the ball and stick representation together with an 'atoms-in-molecules' analysis. The colored lines are bond paths and the small red spheres are bond critical points (bcp's). The bcp's involving the O atoms of dme are highlighted and the density (ρ) is given in arbitrary units $\cdot 10^2$. Stabilizing energies are given in kcal·mol^{-1}. See also entries 1a–6a in Table 1 for a numerical overview and an energy decomposition analysis. See Figure S1 for the values of all intermolecular bcp's. See Figure S2 and Figure S3 for similar graphics involving, respectively, dimethyl thioether (dmt, entries 1b–6b in Table 1) and reference adducts of dme/dmt and hydrogen-/halogen-bond donors (entries 9–12 in Table 1).

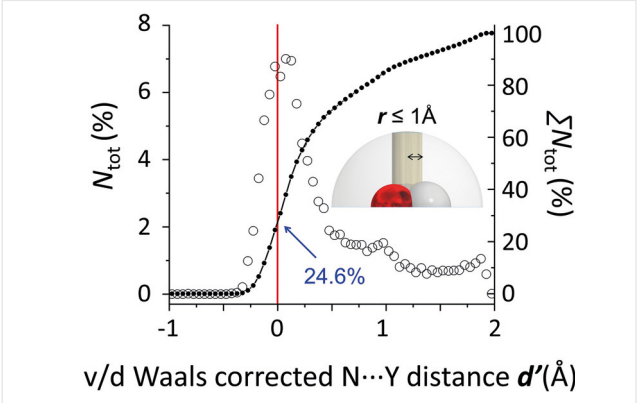


Figure 4 Regular (empty spheres, left y-axis) and cumulative (black spheres, right y-axis) hit fractions (in %) as a function of the van der Waals corrected N···Y distance ***d′*** (in Å) for the datasets where Y is an electron rich atom, ***d*** ≤5Å and ***r*** ≤1Å (i.e., the golden body in the inset figure). N = 9126 within 3893 CIFs and N(***d′*** ≤0) = 2244 within 1460 CIFs (of which Y = O (83.9%), N (13.4%), F (1.2%), Cl (0.9%), Br (0.4%), S (0.2%)). See the methods section for details and Figure S4 for a similar plot of data where Y is the H-atom of any methyl.

and varies between about 50–60%. Orbital interactions are always second with about 20–35% and dispersion always contributes least (~10–25%). The O/S···H/I bcp's (0.011–0.030 a.u., see also Figure S3) are typically denser than those observed in π-hole adducts with **1–6**.

These calculations thus imply that π-hole bonding complexes are comparable in strength to moderate hydrogen-bonding interactions. The physical origins of their stability lies mainly with electrostatics and dispersion. This latter is unsurprising as London dispersion is proportional to the polarization volume of an adduct.[40]

3.2 Analysis of the Cambridge Structure Database (CSD)

The distribution of data within the CSD of electron rich atoms (i.e., Y = ElR) above/below an X–NO_2 moiety with ***d*** ≤5Å and ***r*** ≤1Å is shown in Figure 4 (see methods section and Figure 1 for details, X ≠ aliphatic C, aromatic C_6, nitrate ester or O^-; i.e., no nitrate anions). The data in these N(***d′***) plots are located within the golden body of the inset figure of Figure 4 and represents 9126 intermolecular N···Y distances found within 3893 Crystallographic Information Files (CIFs). The data are plotted as a relative frequency (empty spheres, left y-axis) and a cumulative fraction (black spheres, right y-axis). Both plotting methods clearly reveal a sharp feature near ***d′*** ≈ 0.1 Å and 24.6% (2244 examples within 1460 CIFs) of all these N···Y distances is below the sum of the van der Waals radii of N and Y.[25] The majority of these data involve Y = O (83.9%). Reproducing this search but considering Y = the H-atom of any methyl group resulted in featureless N(***d′***) plots without any instances of overlapping N···H van der Waals shells (see Figure S4). These data thus imply that the interaction is directional in the crystalline state and comparable to interactions with nitrate esters,[11b] small cycloalkane $C_2(CN)_4$ structures,[5d] and $M(CO)_3$ complexes.[7c]

The CSD was further inspected specifically for structures derived from **1–6**.[41] An overview of all 385 instances found with ***d′*** ≤0Å is listed in Table S2 (see SI) and three examples are highlighted in Figure 5. Within structure CEDVEO,[42] a nitrofuran molecule is stacking with itself via reciprocal $^{(C7=O4)}$carbonyl···nitro$^{(N1=O3)}$ interactions (Figure 5a). The intermolecular N1···O4 distance is 0.091Å shorter than the sum of the van der Waals radii of N (1.55Å) and O (1.52Å). A single point energy calculation using the

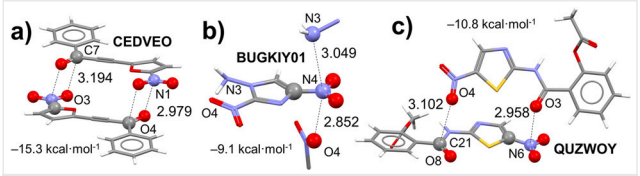


Figure 5 Perspective views of nitro-containing molecules found in the CSD that display a short contact distance with the N-atom of nitro-moiety of: **a**) the nitrofuran derivative found in CEDVEO;[42] **b**) the nitroimidazole derivative found in BUGKIY01;[43] and **c**) the nitrothiazole derivative Nitazoxanide found in QUZWOY.[44] Indicated interaction energies were computed using the crystal structure coordinates at the DFT/B3LYP-D3/def2-TZ2P level of theory.

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coordinates form the crystal structure gave an interaction energy of –15.3 kcal·mol^{-1}, which suggests that each π-interaction is about –7.65 kcal·mol^{-1}.

The nitroimidazole molecule present in BUGKIY01[43] (Figure 5b) is stacked in such a manner within its crystal that one nitro moiety (N4) is sandwiched in between its amine (N3) and the O-atom of the other nitro moiety (O4). The N3···N4 distance (3.049Å) and the N4···O4 distance (2.852Å) are within the van der Waals benchmark. A single point calculation of the molecular trimer indicated an interaction energy of –11.5 kcal·mol^{-1} and calculations of the separate bimolecular adducts gave a nearly identical energy for the amine-nitro (–5.9 kcal·mol^{-1}) and the nitro-nitro interaction (–5.7 kcal·mol^{-1}). Two Nitazoxanide molecules (see also entry **5**.6 in Table S1)[38] are found in structure QUZWOY[44] (Figure 5c) where they are stacked by two amide···nitro interactions. The O3···N6 distance is 0.112 Å shorter than the van der Waals benchmark (1.55 + 1.52 = 3.07Å) and the dimer was computed to be energetically favorable by –10.4 kcal·mol^{-1} (suggesting about –5.2 kcal·mol^{-1} per interaction).

Energy decomposition analyses of the above computations revealed that the adducts are held together mainly by electrostatics (40–50%) and dispersion (40%). This, together with the energy estimated per π-hole interaction (–5 to 7 kcal·mol^{-1}), is in line with the calculations of models nitrofuran **1**, nitroimidazole **3** and nitrothiazole **5** (see Table 1).

3.3 Analysis of the Protein Data Bank (PDB)

The PDB was also inspected for nitro-containing ligands with structures such as **1**–**6** and an overview of the 22 unique ligands found within 27 structures is provided in Table S3 (see SI). This obviously provides too small a dataset for any meaningful statistical evaluation (contrary to an earlier study with nitroaromatics).[12] Manual inspection of the binding pockets in most cases revealed that the nitro moiety is in close proximity to some electron rich entity (i.e., lone-pair or π-electrons). To illustrate this, Figure 6 provides four examples that span general structures **1**–**6**. The nitrofuran ligand **N2F**[45] found within structure 4onm[46] is shown in Figure 6a. **N2F** is covalently bound to cysteine-87 and the NO_2 moiety is π-π stacked with the amide of asparganine-79 with O···N and O···C distances of 3.48Å. The nitro moiety is also H-bonded to the NH_2-moeity of asparganine-123 (not shown), which likely amplifies the π-hole. As is depicted in Figure 6b, the nitroimidazole ligand **2VQ**[47] is anchored within 4m08[48] by coordination to a Zn(histidine)$_3$ complex. The nitro group of **2VQ** is in close proximity to the alcohol O-atom of threonine-200 with a short N···O distance of 3.14 Å. Within the (*Homo sapiens*) aldose kinase structure 3v35[49] shown in Figure 6c, a dimethylformamide molecule (DMF-318) is stacked on top of the nitro group of nitrothiazole ligand **NTI**[50] (Nitazoxanide, see also entry **5**.6 in Table S1 and Table S3)[38] with a N···N distance of 3.32 Å.

Interestingly the pentameric structures of acetylcholine binding protein 3wtl and its mutant 3wtm[51] contain a total of ten nitroethene diamine derived ligands **N1Y**[50b,52] (Nitenpyram, see also entry **6**.3 in Table S1 and Table S3). As is illustrated in Figure 6d, these ligands are located in between the chains and π-π stacked on top of tyrosine-185 with an average distance of 3.85Å between the centroids of the rings in **N1Y** (white sphere) and the tyrosine-185 rings (green sphere). Additionally, the S-atom of methionine-144 is consistently in close proximity to the nitro N-atom of **N1Y** with an average S···N distance of 3.50 Å. While this distance is somewhat larger than the sum of the van der Waals radii of N (1.55) and S (1.80), that is, 3.35 Å, it has been documented that van der Waals overlap is not a strict prerequisite for π-hole interactions with S-lone pairs.[12,57] Nearly identical interactions were observed with ligand **IM4**[58] (a structural analogue of **NIY**) within structures 2xju[59] and 3wth[51] (see also Table S3). Ligand **IM4** is known as imidacloprid and a widely used neonicotinoid insecticide.[60]

While not shown in Figure 6, in all instances highlighted above the nitro moiety was also hydrogen bonded by some residue. That is: to asparagine-123 in 4onm, to histidine-64 in 4m08, to tryptophan-111 in 3v35, and to serine-186 in

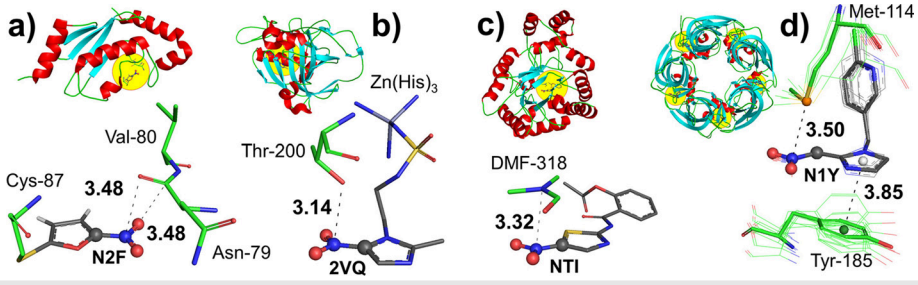


Figure 6 Perspective views of nitro-containing ligands found in the PDB and the residues that display a short contact distance with the N-atom of the ligand's nitro moiety: **a**) *Homo sapiens* ubiquitin-conjugating enzyme E2 variant 2 (4onm)[46] with nitrofuran ligand **N2F**;[45] **b**) Structure 4m08[48] is a mutant of *Nitrospira defluvii* chlorite dismutase containing nitroimidazole ligand **2VQ**;[47] **c**) *Homo sapiens* aldose kinase structure 3v35[49] contains nitrothiazole ligand **NTI**[50a] (nitazoxanide);[38] **d**) overlay of (*Lymnaea stagnalis*) acetylcholine binding protein structures within 3wtl and 3wtm[51] containing a total of ten nitroethene diamine-derived ligands **N1Y**.[50b,52] See text for more details and see Figure S5 for similar structures involving a nitro-aliphatic ligand[53] in **8acn**[54] and a nitro-ester ligand[55] in **5e29**.[56]

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3wtl and 3wtm. This hydrogen bonding will likely amplify the π-hole present on the ligands and thus lead to stronger interaction energies, as illustrated by the MEPs in Figure 1 and demonstrated elsewhere for other nitro-containing compounds.[12–14] It is obviously unlikely that the π-hole interactions described above are the main driver behind the binding of these various ligands within their protein pockets. Especially as (charge-assisted) hydrogen-bonding interactions can be much larger than the approximately –5 kcal·mol^{-1} estimated for a single π-hole interaction (see Table 1). It is still remarkable that in the few protein structures containing **1**–**6**-like ligands, many involve short distances between the nitro moiety and an electron rich part of a residue. It has furthermore been highlighted in a recent review that the nitro group in nitroimidazole drugs is often essential for a high inhibition.[61] Besides, the stabilizing energies computed in model systems are not that low and comparable to hydrogen bonding in the water dimer (–4.7 kcal·mol^{-1}).[62] Such weak interactions are well known to be relevant to the structure and/or function of both small and large molecular assemblies. For example: H-π interactions can be directional,[63] such as in the crystal packing of indolocarbazoles;[64] the α-H of amino acid side chains are involved in weak H-bonding interactions;[65] weak S···π$^+$ interactions have been invoked in biocatalysis;[57] π-hole interactions between carbonyls[66] are known to direct protein folding;[7b] and weak C–H···π interactions with carbohydrates are preserved in protein pockets[67] and exploited in artificial receptors.[68]

4 Summary and Conclusions

In summary, many small nitro-containing molecules are widely used as drugs including Nitrofurantoin,[32] Nifurtimox,[33] Fexinidazole,[34] Metronidazole,[35] Delamanid,[36] Azathioprine,[37] Benznidazole,[33] Nitazoxanide[38] and Ranitidine.[39] These medicine can be classified according to the general structure of their nitro-bearing moieties **1**–**6**. MEPs of **1**–**6** predict the presence of a π-hole on the nitro groups (Figure 1). Model calculations show that π-hole adduct with **1**–**6** are mainly driven by electrostatic attraction and dispersion and that interaction energies can be as large as typical hydrogen- or halogen-bonded adducts (–6.5 kcal·mol^{-1}, see Table 1 and Figure 2). Within the crystal structures of small molecules, electron-rich atoms cluster near the N-atom of XNO_2 entities and about 25% of all data within a volume characterized by ***d*** ≤5Å and ***r*** ≤1Å displays van der Waals overlap (Figure 4). These structures include many compounds derived from **1**–**6**, as highlighted for three structures in Figure 5, with estimated interaction energies in line with the model calculations. A PDB survey revealed similar short distances involving the nitro-moiety of various ligands (Figure 6).

It is thus concluded that intermolecular π-hole interactions with nitro compounds such as **1**–**6** are similar in strength as a typical hydrogen and halogen bond at –3.9 to –6.5 kcal·mol^{-1}. These π-hole interactions are a directional force in small molecule crystals and also occur in ligand–protein complexes. These findings might be relevant to understand and manipulate ligand–protein complexation phenomena with nitro-bearing compounds with the ultimate goal of improving inhibitory medicine.

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Supporting Information

Supporting information for this article is available online at https://doi.org/10.1055/s-0039-1690209.

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